

# ShellSentry – Secure Natural Language to Bash Execution

## Overview

ShellSentry is a secure AI-powered platform that converts natural language requests into validated Bash commands and executes them on remote Linux systems. The platform was developed as a senior cybersecurity project to improve Linux administration efficiency while maintaining strong security controls.

## Problem

AI models can generate Bash commands from natural language, but executing these commands directly on production systems can introduce significant security risks, including destructive commands, prompt injection attacks, and privilege abuse.

## Solution

ShellSentry applies a defense-in-depth approach that treats all AI-generated commands as untrusted. Commands pass through multiple validation layers before execution, ensuring secure automation across remote Linux servers.

## Key Features

- Natural Language to Bash conversion
- Retrieval-Augmented Generation (RAG)
- Multi-host SSH execution
- Command validation and filtering
- Audit logging and reporting
- Script archiving and scheduling
- Host-aware command generation

## Technologies Used

- Python
- Flask
- Paramiko
- FAISS
- SQLite
- OpenAI-Compatible LLM APIs
- HTML/CSS/JavaScript

## My Contributions

- Designed the system architecture and security workflow
- Implemented command validation mechanisms

- Integrated RAG-based retrieval
- Developed SSH orchestration and audit logging
- Conducted security and performance testing

## **Outcome**

The project demonstrated secure AI-assisted Linux administration while preventing unsafe command execution through layered security controls.